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Changes in antimicrobial resistance of *Escherichia coli* isolated from community-associated urinary tract infection before and during the COVID-19 pandemic in India



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ABSTRACT

Purpose: The impact of the COVID-19 pandemic on antimicrobial resistance (AMR) is largely studied in healthcare settings. There is a need to understand the fluctuations in AMR during pandemic at the community level. With urinary tract infection (UTI) being one of the most common infections in the community, the AMR profile of community-acquired UTI (CA-UTI) is considered representative AMR at the community level.

Methods: The study was taken in a cohort of patients with a clinical diagnosis of CA-UTI. The four study sites represented different community health centres in India. *Escherichia coli* isolates were analysed phenotypically and genotypically for AMR pre-COVID (October 2019–February 2020) and in the first (March 2020–February 2021) and second waves of COVID-19 (March 2021–December 2021).

Results: *E. coli* was the predominant uropathogen (229, 82%). Increased susceptibility to nitrofurantoin was observed during the pandemic. Reduced susceptibility to first-line oral antibiotics and carbapenems was seen during the second wave, and an increased minimum inhibitory concentration (MIC₅₀) to beta-lactams and fluoroquinolones was seen during the pandemic. Genomic analysis of *E. coli* isolates showed some AMR genes (*aacC1*, *aacC4*, *SHV*, *QepA*) only during the second wave.

Conclusion: One good outcome of the pandemic was increased susceptibility to nitrofurantoin, while drawback was a significant decrease in susceptibility to oral antibiotics during the second wave and increased MIC₅₀ of some antibiotics. Decreased susceptibility to last-resort carbapenems and the occurrence of various AMR genes during the second wave of the pandemic are of great concern.

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1. Introduction

The COVID-19 pandemic has had a manifold effect on antimicrobial resistance (AMR), which is a bigger threat than the pandemic itself [1,2]. It is important to discern the impact of the COVID-19 pandemic on AMR in the community.

2. Materials and methods

2.1. Study site, period, and participants

The study was undertaken as part of an ongoing study, Community-Acquired UTI and Emerging Drug Resistance (CAUTION-ED), to determine epidemiology and antimicrobial susceptibility among CA-UTI cases in the outpatient department of community health centres of four regions of India. Patients aged >1.5 years suspected of CA-UTI recruited in the main study were included in the present study, and the phenotypic and genotypic AMR of *E. coli* from urine samples was analysed pre-COVID (October 2019–February 2020) and during the first wave of COVID-19 (March 2020–February 2021) and second wave (March 2021–December 2021). In March 2020, WHO declared the COVID-19 pandemic; in India, the first wave lasted until February 2021, after which the delta variant led to the second wave until December 2021 [3].

2.2. Detection of phenotypic AMR

The growth of a single type of organism with a colony count $\geq 10^3$ (CFU/mL) from midstream urine was considered significant. Identification was completed by MALDI-TOF MS (bioMérieux, Nürtingen, Germany), AST was performed by VITEK-2 (bioMérieux, Durham, NC, USA), and MIC₅₀ was calculated.

2.3. Genotypic analysis for AMR genes

Due to cost constraints, only 12 isolates were taken up for genotypic analysis (three randomly selected *E. coli* isolates, representing each study period from four centres). A microbial DNA qPCR assay was performed using a microbial DNA qPCR array plate (24-array, BAID-1901ZRD-24; QIAGEN, Hilden, Germany) to detect bacterial 16s rRNA gene sequences and AMR genes.

2.4. Statistical analysis

Statistical analysis was performed using Stata 16.0 (StataCorp, College Station, TX, USA). AST data were compared between various study periods using a χ^2 test. MIC₅₀ data were analysed, and temporal MIC₅₀ trends were examined using a Cochran–Armitage test [4]. GraphPad Prism version 8.0.0 for NOVA (GraphPad Software, San Diego, CA, USA), followed by a Dunnett's multiple comparison test, was used for AMR gene analysis.

3. Results

3.1. Patient and bacterial profile

A total of 2991 patients were enrolled during the entire study period. Of those patients, 331 (11.06%) had significant bacteriuria; in 229 (82%) patients, the predominant pathogen was *E. coli*, and in 50 (18%), it was *K. pneumoniae*.

3.2. Phenotypic AMR data

There was a statistically significant decrease in susceptibility to amoxicillin-clavulanic acid (42% vs 58%), ampicillin (22% vs 37%), and co-trimoxazole (36% vs 56%) during the second wave compared with the first wave, and a statistically significant increase in susceptibility to nitrofurantoin (87% vs 71%) during the second wave in comparison with the first wave and pre-COVID period (Supplementary Table 1). An increase in MIC₅₀ for amikacin ($P = 0.0112$; <0.05) was found during the second wave compared with pre-COVID.

3.3. Genotypic data

Genotyping showed the presence of AMR genes to aminoglycosides, beta-lactams (including carbapenems), fluoroquinolones, tetracyclines, and macrolides, during pre-COVID and at the time of first and second COVID-19 waves, and their frequency is depicted in Figure 1. Though AMR genes were found to be present heterogeneously among the tested isolates, the occurrence of some AMR genes only during second wave cannot be ignored (Supplementary Table 2).

4. Discussion

This study focused on *E. coli* from CA-UTI patients to evaluate the changes in AMR in the community during the pandemic. Only nitrofurantoin in WHO AWaRe access category fared better [5]. Similar studies in the community have not been performed, and the best available data for comparison come from hospital settings. The ICMR Antimicrobial Resistance Research and Surveillance Network (ICMR-AMRSN) found decreased susceptibility to oral antibiotics but no significant difference during the pandemic; this may be due to lower usage of oral antibiotics in hospital settings. Fosfomycin remained 100% susceptible; 97–93% susceptibility is reported in hospital settings [6,7]. Awaiting culture and sensitivity reports, for uncomplicated CA-UTI, fosfomycin and nitrofurantoin should be considered as empirical antibiotics.

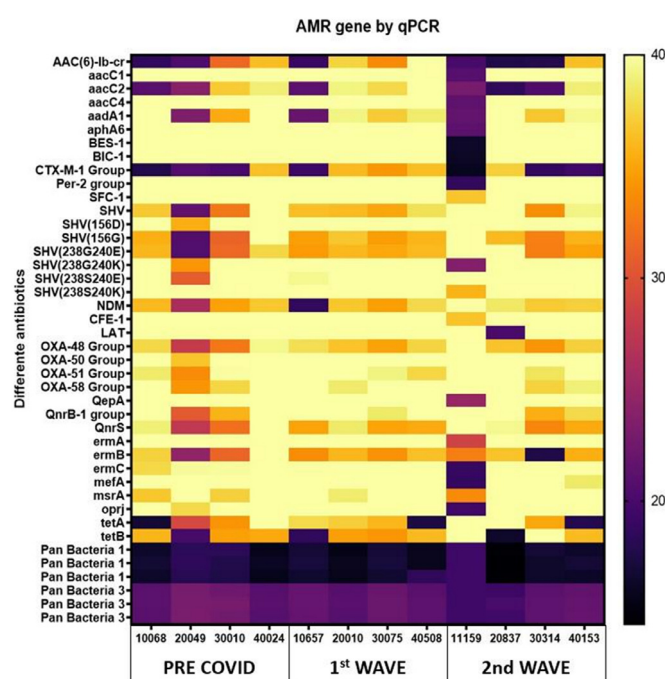


Figure 1. Heat map of AMR genes before and during COVID-19. The figure represents the distribution of AMR genes in *E. coli* detected via qPCR. The gradient of colour from yellow to dark blue represents more copies of a particular gene. *E. coli* isolates from Delhi (North India – Centre 1) were 10068, 10657, and 11159. *E. coli* isolates from Bhuvaneshwar (East India – Centre 2) were 20049, 20010, and 20837. *E. coli* isolates from Jodhpur (West India – Centre 3) were 30100, 30075, and 30314. *E. coli* isolates from Bengaluru (South India – Centre 4) were 40024, 40508, and 40153.

Additionally, NDM beta-lactamases were found during entire study period. Similarly, 25% of the *E. coli* from hospital setting had NDM-1 during the first wave, which increased to 31% in the second wave [6,7]. This is of great concern, as the molecular mechanism indicates impending decreased susceptibility to last-resort injectables even at the community level.

In conclusion, this study highlights notable disruption in AMR during the pandemic and the need to extend AMR surveillance undertaken in hospitals to community. The changing AMR pattern can be attributed to the indiscriminate use of antibiotics to treat COVID-19 patients, emphasizing the need for antimicrobial stewardship even during pandemics. In India, national guidelines for CA-UTI should consider fosfomycin and nitrofurantoin as empirical choices.

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Competing interests: None declared.

Ethical approval: Ethical approval for this study was granted (IEC-192/05.04.2019, RP-28/2019) by the Institute Ethics Committee, AI-IMS, New Delhi (registration no. ECR/538/Inst/DL/2014 with Directorate General of Health Services).

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2024.02.022](https://doi.org/10.1016/j.jgar.2024.02.022).

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